



Divyadarshan Soni
Chemical Engineering
Indian Institute of Technology Bombay

22B0387
B.Tech.
Gender: Male
DOB: 26/01/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	Core: 183 Total: 201

SCHOLASTIC ACHIEVEMENTS

- Secured an impressive percentile of **97.60** in JEE Adv with rank **AIR 3730** among **0.16** million aspirants('22)
- Secured an impressive percentile of **99.25** in JEE Main with rank **AIR 6891** among **1.02** million aspirants('22)
- Secured **9 SPI** in the sixth semester and grabbed **AA/AB** grades in **9** courses for academic excellence(May'25)

PROFESSIONAL EXPERIENCE

AI Engineering Intern | *Smart-Analytica*

(May'25 - July'25)

Awarded Letter of Recommendation by CTO at Smart-Analytica for exceptional work & exemplary performance

- Engineered advanced AI assistant unifying Chatbot, RCA, and reporting to streamline Big Data workflows
- Efficiently leveraged Neo4j Graph DB and Milvus Vector DB to store and retrieve over 1M data records daily

Chatbot	- Engineered 5+ prompting methods improving LLM responses, increasing accuracy by over 20% - Designed 7 robust metadata filters and optimized semantic search in Milvus through Python - Achieved 80% overall response accuracy with approx 12 seconds for end-to-end query processing
RCA	- Created ranking module using 4 scoring algorithms to select top tests from 15+ available diagnostics - Automated dynamic execution of pre-written diagnostic tests, reducing manual troubleshooting time - Linked 30+ root causes to both hard-coded and RAG-derived solutions for instant RCA reporting

TECHNICAL PROJECTS

Digital Diagnosis of Obstructive Sleep Apnea | *Academic Project* | Prof. Saket k. Choudhary (Aug'24 - Dec'24)

Developed automated pipeline to detect Sleep Apnea events from polysomnography data, improving diagnostic efficiency

- Processed **32-channel** EDF files in R language, extracting key signals (Flow.Patient, SpO₂) for the analysis
- Designed signal processing algorithms to detect the apnea and hypopnea events with more than **90%** accuracy
- Applied **PCA** and correlation analysis across all **32** data channels to identify the strongest predictive features

Protein-Nucleotide-Structure Conservation Study | *Course Project* | Prof. Saket k. Choudhary (Aug'24 - Dec'24)

Studied evolutionary conservation of protein, nucleotide, and structure across species to reveal cross-domain relationship

- Aligned sequences of **50** genes across **5** species using Biopython, generating over **250** conservation profiles
- Integrated AlphaFold predictions with Blosum62 scores to map cross-domain conservation for all proteins
- Performed correlation analysis between conservation types, revealing the significant inter-domain relationships

Computer Vision-based Book Angle Detection | *Self Project* | Image Processing (Oct'23 - Nov'23)

Developed a vision system to detect books and calculate their orientation angles using YOLOv8 and image processing

- Trained a custom YOLOv8 object detection model, achieving **92%** mAP on **500+** precisely labeled images
- Applied Gaussian Blur for noise reduction and implemented line pruning techniques to refine detection accuracy
- Implemented angle estimation algorithms with trigonometric calculations, achieving $\pm 2^\circ$ average relative error

Rainfall Prediction Using Machine Learning Models | *Self Project* | Machine Learning (Jun'24 - Jul'24)

Designed end-to-end ML system to forecast rainfall using multiple algorithms with optimized preprocessing and evaluation

- Analyzed **145k+** records, effectively handling **10%+** missing data and **25%** class imbalance via preprocessing
- Compared **7** models: [RF, LR, DT, NN, LGBM, CatBoost, XGBoost] to identify the best rainfall predictor

TECHNICAL SKILLS

Languages	Python R C++ C SQL MATLAB L ^A T _E X Cypher
Software	OpenRouter VsCode RStudio Excel MySQL Figma Fusion360 ASPEN DWSIM
Python Libraries	NumPy Pandas sklearn pytorch OpenCV Matplotlib Plotly transformers h2o
Miscellaneous	ChromaDB Milvus Neo4J DB2(IBM) Ms365 Overleaf Da Vinci Photopea Gimp

EXTRA-CURRICULARS

Culturals	- Trained with the National Cadet Corps for 9+ months gaining professional army education ('23) - Proficient in playing Guitar, Cajon and Keyboard with more than 8 years of experience (Present)
Technical	- Dedicated 60+ hours in studying, understanding and practising DSA through Udemy (Aug'25) - Achieved a rating of 1600+ on CodeChef by competing in competitive coding contests (Present) - Earned Machine Learning certification, mastering algorithms and practical implementations (Aug'24)